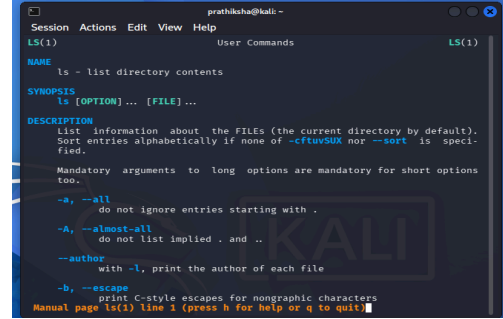
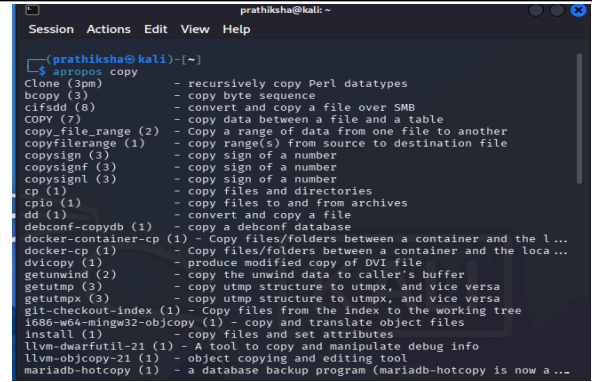
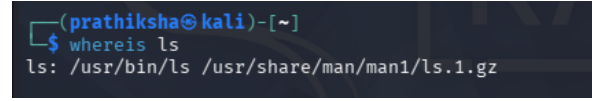


LOS Assignment - 2

1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man <command_name>	man ls	
To get manual page for the unknown command	apropos	apropos <keyword>	apropos copy	
To know the source file binary	whereis	whereis <command_name>	whereis ls	

To know the path of the command	which	which <command_name>	which ls	<pre>(prathiksha@kali)-[~] \$ which ls ls: aliased to ls --color=auto</pre>
To know the command is external or internal	type	type <command_name>	type cd	<pre>(prathiksha@kali)-[~] \$ type cd cd is a shell builtin</pre>
To get help for the internal command	help	help <command_name>	help cd	<pre>(prathiksha@kali)-[~] \$ help cd cd: cd [-L -P [-e]] [-@] [dir] Change the shell working directory. Change the current directory to DIR. The default DIR is the value of the HOME shell variable. If DIR is "~", it is converted to \$OLDPWD. The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory. If DIR begin with a slash (/), then CDPATH is not used. If the directory is not found, and the shell option 'cdable_vars' is set, the word is assumed to be a variable name. If that variable has a value its value is used for DIR. Options: -L force symbolic links to be followed: resolve symbolic links in DIR after processing instances of '..' -P use the physical directory structure without following symbolic links: resolve symbolic links in DIR before processing instances of '..' -e if the -P option is supplied, and the current working directory cannot be determined successfully, exit with a non-zero status</pre>
To list out bash commands	compgen	compgen -b	compgen -b	<pre>(prathiksha@kali)-[~] \$ compgen -b _ : _ alias bg bind break builtin caller cd command compgen complete compspt continue declare dirs disown echo enable eval exec exit export false</pre>

To know the usage of the command	--help	<command_name> --help	ls --help	<pre>(prathiksha@kali)-[~] └─\$ ls --help Usage: ls [OPTION]... [FILE]... List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified. Mandatory arguments to long options are mandatory for short options too. -a, --all do not ignore entries starting with . -A, --almost-all do not list implied . and .. --author with -l, print the author of each file -b, --escape print C-style escapes for nongraphic characters --block-size=SIZE with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below -B, --ignore-backups do not list implied entries ending with ~ -C with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first</pre>
----------------------------------	--------	--------------------------	-----------	--

2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date	date	<pre>(prathiksha@kali)-[~] └─\$ date Thursday 20 August 2026 08:08:38 PM IST</pre>
To print calendar	cal	cal	cal	<pre>(prathiksha@kali)-[~] └─\$ cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31</pre>
To print kernel version	uname	uname -r	uname -r	<pre>(prathiksha@kali)-[~] └─\$ uname -r 6.19.14+kali-amd64</pre>
To print default shell	echo	echo \$SHELL	echo \$SHELL	<pre>(prathiksha@kali)-[~] └─\$ echo \$SHELL /usr/bin/zsh</pre>
To print currently logged in user	whoami	whoami	whoami	<pre>(prathiksha@kali)-[~] └─\$ whoami prathiksha</pre>

To create shortcut for command	alias	alias name= 'command'	alias ll='ls -l'	<pre>(prathiksha@kali)-[~] \$ alias ll='ls -l' (prathiksha@kali)-[~] \$ alias alias diff='diff --color=auto' alias egrep='egrep --color=auto' alias fgrep='fgrep --color=auto' alias grep='grep --color=auto' alias ip='ip --color=auto' alias l='ls -CF' alias la='ls -A' alias ll='ls -l' alias ls='ls --color=auto'</pre>
To delete shortcut	unalias	unalias name	unalias ll	<pre>(prathiksha@kali)-[~] \$ unalias ll (prathiksha@kali)-[~] \$ alias ll bash: alias: ll: not found (prathiksha@kali)-[~] \$</pre>
To change the timestamp of the file	touch	touch <filename>	touch a.txt	<pre>(prathiksha@kali)-[~] \$ touch a.txt</pre>
To clear the screen	clear	clear	clear	<pre>(prathiksha@kali)-[~] \$</pre>
To create empty files	touch	touch <file1> <file2>	touch a.txt b.txt	<pre>(prathiksha@kali)-[~] \$ touch a.txt b.txt</pre>
To know disk usage	du	du -sh <path>	du -sh /home	<pre>(prathiksha@kali)-[~] \$ du -sh /home 236M /home</pre>

To know free space in the system	df	df -h	df -h	<pre>(prathiksha@kali)-[~] \$ df -h Filesystem Size Used Avail Use% Mounted on udev 951M 0 951M 0% /dev tmpfs 197M 100K 196M 1% /run /dev/sda1 19G 17G 977M 95% / tmpfs 985M 4.0K 985M 1% /dev/shm tmpfs 985M 8.0K 985M 1% /tmp none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service tmpfs 197M 112K 197M 1% /run/user/1000 none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.servi ce</pre>
To know about the Linux release	cat	cat /etc/os-release	cat /etc/os-release	<pre>(prathiksha@kali)-[~] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.2" VERSION="2026.2" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"</pre>
To show login history	last	last	last	<pre>(prathiksha@kali)-[~] \$ last prathiks tty? :0 Thu Aug 20 19:44 - still logged in prathiks tty? :0 Wed Aug 19 00:00 - still logged in prathiks tty? :0 Tue Aug 18 15:00 - still logged in prathiks tty? :0 Mon Aug 17 22:00 - still logged in prathiks tty? :0 Mon Aug 17 20:36 - still logged in prathiks tty? :0 Mon Aug 17 20:00 - still logged in prathiks tty? :0 Sun Aug 16 22:46 - still logged in prathiks tty? :0 Sat Aug 15 17:43 - still logged in prathiks tty? :0 Sat Aug 15 17:31 - still logged in prathiks tty? :0 Sat Aug 15 10:06 - still logged in prathiks tty? :0 Sat Aug 15 09:34 - still logged in prathiks tty? :0 Sat Aug 15 09:10 - still logged in prathiks tty? :0 Fri Aug 14 21:50 - still logged in prathiks tty? :0 Fri Aug 14 21:06 - still logged in prathiks tty? :0 Fri Aug 14 20:30 - still logged in prathiks tty? :0 Fri Aug 14 19:39 - still logged in prathiks tty? :0 Wed Aug 12 22:25 - still logged in prathiks tty? :0 Tue Aug 11 14:00 - still logged in prathiks tty? :0 Tue Aug 11 14:01 - still logged in prathiks tty? :0 Tue Aug 11 10:16 - still logged in utmpdb begins Tue Aug 11 10:16:41 2026</pre>
To check the current user privileges	sudo	sudo -l	sudo -l	<pre>(prathiksha@kali)-[~] \$ sudo -l Matching Defaults entries for prathiksha on kali: secure_paths=/usr/sbin:/usr/bin/:/usr/bin/:/usr/sbin/:/bin Runas and Command-specific defaults for prathiksha: Defaults!/usr/sbin/sudo env_keep+="SUDO_EDITOR EDITOR VISUAL" User prathiksha may run the following commands on kali: (ALL : ALL) ALL</pre>
To find about the Linux release	uname	uname -a	uname -a	<pre>(prathiksha@kali)-[~] \$ uname -a Linux kali 6.19.14-kali-amd64 #1 SMP PREEMPT_DYNAMIC kali 6.19.14-kali (20 26-05-05) x86_64 GNU/Linux</pre>
To check system uptime	uptime	uptime	uptime	<pre>(prathiksha@kali)-[~] \$ uptime 21:00:42 up 1:16, 1 user, load average: 0.03, 0.09, 0.18</pre>
To shutdown	shutdown	shutdown [time]	shutdown now	<pre>(prathiksha@kali)-[~] \$ shutdown now</pre>

To restart	reboot	reboot	reboot	<pre>(prathiksha@kali)-[~] \$ reboot</pre>
------------	--------	--------	--------	--

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd	cd	<pre>(prathiksha@kali)-[~] \$ cd</pre>
To navigate to the parent directory	cd	cd ..	<pre>(prathiksha@kali)-[~] \$ cd .. (prathiksha@kali)-[/home] \$</pre>
To navigate to the child directory	cd	cd <dirname>	<pre>(prathiksha@kali)-[~] \$ cd practice (prathiksha@kali)-[~/practice] \$</pre>
Alternate command to cd	pushd	pushd <dirname>	<pre>(prathiksha@kali)-[~] \$ pushd practice ~/practice ~ (prathiksha@kali)-[~/practice] \$</pre>
To go back to the previous directory	cd -	cd -	<pre>(prathiksha@kali)-[~/practice] \$ cd - ~</pre>
To go to the root directory	cd /	cd /	<pre>(prathiksha@kali)-[~] \$ cd / (prathiksha@kali)-[/] \$</pre>

4. File System

Task	Syntax	Command
How to identify the file system	df -T	<pre>(prathiksha@kali)-[~] \$ df -T Filesystem Type 1K-blocks Used Available Use% Mounted on udev devtmpfs 972960 0 972960 0% /dev tmpfs tmpfs 201688 1008 200680 1% /run /dev/sda1 ext4 19353424 17353676 991316 95% / tmpfs tmpfs 1008432 4 1008428 1% /dev/shm tmpfs tmpfs 1008432 52 1008380 1% /tmp none tmpfs 1024 0 1024 0% /run/credentials/ge tty@tty1.service tmpfs tmpfs 201684 112 201572 1% /run/user/1000 none tmpfs 1024 0 1024 0% /run/credentials/sy stemd-journald.service (prathiksha@kali)-[/] \$ df -T / Filesystem Type 1K-blocks Used Available Use% Mounted on /dev/sda1 ext4 19353424 17353676 991316 95% /</pre>

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use **globbing patterns wherever possible**, rather than specifying filenames individually.

```
(prathiksha@kali)-[~]
$ mkdir practice11
(prathiksha@kali)-[~]
$ cd practice11
(prathiksha@kali)-[~/practice11]
$ touch index.html login.html admin.html report.txt access.log.error.log us
er1.txt user2.log file123 backup1.html backup2.txt
```

```
(prathiksha@kali)-[~/practice11]
$ shopt -s extglob
(prathiksha@kali)-[~/practice11]
$ shopt extglob
extglob on
```

a. List all files ending in .html.

```
(prathiksha@kali)-[~/practice11]
$ ls *.html
admin.html backup1.html index.html login.html
```

- b. List all files **other than .html files** using extended globbing.

```
(prathiksha@kali)~/practice11
$ ls !(*.html)
access.log.error.log  backup2.txt  file123  report.txt  user1.txt  user2.log
```

- c. List filenames containing at least one numeric digit.

```
(prathiksha@kali)~/practice11
$ ls *[[0-9]]*
backup1.html  backup2.txt  file123  user1.txt  user2.log
```

- d. List files whose names begin with user followed by a single digit.

```
(prathiksha@kali)~/practice11
$ ls user[0-9]*
user1.txt  user2.log
```

- e. List files ending in either .txt or .log using a single **extended glob pattern**.

```
(prathiksha@kali)~/practice11
$ ls *.@(txt|log)
access.log.error.log  backup2.txt  report.txt  user1.txt  user2.log
```

- f. List files that do **not** end in .txt or .log.

```
(prathiksha@kali)~/practice11
$ ls !(*.txt|*.log)
admin.html  backup1.html  file123  index.html  login.html
```

- g. List files beginning with either access or error using extended globbing.

```
(prathiksha@kali)~/practice11
$ ls @(access|error)*
access.log.error.log
```

h. List .html files whose filenames contain a numeric digit.

```
(prathiksha@kali)-[~/practice11]
$ ls *[0-9]*.html
backup1.html
```

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

Select all files except notes.txt.

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

```
(prathiksha@kali)-[~]
$ mkdir practice22

(prathiksha@kali)-[~]
$ cd practice22

(prathiksha@kali)-[~/practice22]
$ shopt -s extglob

(prathiksha@kali)-[~/practice22]
$ touch access.log error.log auth.log auth.log.1 auth.log.2 system.log notes.txt

(prathiksha@kali)-[~/practice22]
$ ls @(access|error|auth).log
access.log auth.log error.log

(prathiksha@kali)-[~/practice22]
$ ls !(notes.txt)
access.log auth.log auth.log.1 auth.log.2 error.log system.log

(prathiksha@kali)-[~/practice22]
$ ls auth.log.[12]
auth.log.1 auth.log.2
```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
(prathiksha@kali)~/practice22
$ rm *

(prathiksha@kali)~/practice22
$ touch malware1.log malware2.log malwareA.log malware_report.txt normal.log

(prathiksha@kali)~/practice22
$ ls malware*[0-9]*.log
malware1.log  malware2.log
```

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed_login or successful_login.

Construct a pattern that lists all .log files **except** firewall.log.

```
(prathiksha@kali)~/practice22
$ rm *

(prathiksha@kali)~/practice22
$ touch failed_login.log successful_login.log login_backup.log login.txt firewall.log

(prathiksha@kali)~/practice22
$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log

(prathiksha@kali)~/practice22
$ ls !(firewall).log
failed_login.log  login.txt  login_backup.log  successful_login.log
```

7. Filters

- a. Display the first **10 entries** of login.log.

```
(prathiksha@kali)~/practice22
$ cd
$ cd
$ mkdir practice33
$ cd practice33
$ seq 1 100 | awk '{print "login entry "$1}' > login.log
$ seq 1 100 | awk '{print "access entry "$1}' > access.log
$ seq 1 500 | awk '{print "security event "$1}' > security.log
$ seq 1 500 | awk '{print "incident event "$1}' > incident.log
$ head -10 login.log
login entry 1
login entry 2
login entry 3
login entry 4
login entry 5
login entry 6
login entry 7
login entry 8
login entry 9
login entry 10
```

- b. Display the first **5 entries** of access.log.

```
(prathiksha@kali)~/practice33
$ head -5 access.log
access entry 1
access entry 2
access entry 3
access entry 4
access entry 5
```

- c. Display the first **15 security events** from security.log.

```
(prathiksha@kali)~/practice33
$ head -15 security.log
security event 1
security event 2
security event 3
security event 4
security event 5
security event 6
security event 7
security event 8
security event 9
security event 10
security event 11
security event 12
security event 13
security event 14
security event 15
```

- d. Display the first **20 events** recorded in incident.log.

```
(prathiksha@kali)~/practice33
$ head -20 incident.log
incident event 1
incident event 2
incident event 3
incident event 4
incident event 5
incident event 6
incident event 7
incident event 8
incident event 9
incident event 10
incident event 11
incident event 12
incident event 13
incident event 14
incident event 15
incident event 16
incident event 17
incident event 18
incident event 19
incident event 20
```

- e. Display only the **first entry** in login.log.

```
(prathiksha@kali)~/practice33
$ head -1 login.log
login entry 1
```

- f. Display the first **25 web requests** recorded in access.log.

```
(prathiksha@kali)~/practice33
$ head -25 access.log
access entry 1
access entry 2
access entry 3
access entry 4
access entry 5
access entry 6
access entry 7
access entry 8
access entry 9
access entry 10
access entry 11
access entry 12
access entry 13
access entry 14
access entry 15
access entry 16
access entry 17
access entry 18
access entry 19
access entry 20
access entry 21
access entry 22
access entry 23
access entry 24
access entry 25
```

g. Display the last **10 entries** of login.log.

```
(prathiksha@kali)~/practice33  
$ tail -10 login.log  
login entry 91  
login entry 92  
login entry 93  
login entry 94  
login entry 95  
login entry 96  
login entry 97  
login entry 98  
login entry 99  
login entry 100
```

h. Display the last **5 entries** of access.log.

```
(prathiksha@kali)~/practice33  
$ tail -5 access.log  
access entry 96  
access entry 97  
access entry 98  
access entry 99  
access entry 100
```

i. Display the **20 most recent security events** from security.log.

```
(prathiksha@kali)~/practice33  
$ tail -20 security.log  
security event 481  
security event 482  
security event 483  
security event 484  
security event 485  
security event 486  
security event 487  
security event 488  
security event 489  
security event 490  
security event 491  
security event 492  
security event 493  
security event 494  
security event 495  
security event 496  
security event 497  
security event 498  
security event 499  
security event 500
```

- j. Display the **25 most recent events** from incident.log.

```
(prathiksha@kali)~/practice33
$ tail -25 incident.log
incident event 476
incident event 477
incident event 478
incident event 479
incident event 480
incident event 481
incident event 482
incident event 483
incident event 484
incident event 485
incident event 486
incident event 487
incident event 488
incident event 489
incident event 490
incident event 491
incident event 492
incident event 493
incident event 494
incident event 495
incident event 496
incident event 497
incident event 498
incident event 499
incident event 500
```

- k. Display only the **last entry** in security.log.

```
(prathiksha@kali)~/practice33
$ tail -1 security.log
security event 500
```

- l. Display the last **50 events** recorded in incident.log.

```
(prathiksha@kali)~/practice33
$ tail -50 incident.log
incident event 451
incident event 452
incident event 453
incident event 454
incident event 455
incident event 456
incident event 457
incident event 458
incident event 459
incident event 460
incident event 461
incident event 462
incident event 463
incident event 464
incident event 465
incident event 466
incident event 467
incident event 468
incident event 469
incident event 470
incident event 471
incident event 472
incident event 473
incident event 474
incident event 475
incident event 476
incident event 477
incident event 478
incident event 479
incident event 480
incident event 481
incident event 482
incident event 483
incident event 484
incident event 485
incident event 486
incident event 487
incident event 488
incident event 489
incident event 490
incident event 491
incident event 492
incident event 493
incident event 494
incident event 495
incident event 496
```

m. Display the contents of login.log starting from **line 41 until the end of the file**.

```
(prathiksha@kali) ~/practice33
$ tail -n +41 login.log
login entry 41
login entry 42
login entry 43
login entry 44
login entry 45
login entry 46
login entry 47
login entry 48
login entry 49
login entry 50
login entry 51
login entry 52
login entry 53
login entry 54
login entry 55
login entry 56
login entry 57
login entry 58
login entry 59
login entry 60
login entry 61
login entry 62
login entry 63
login entry 64
login entry 65
login entry 66
login entry 67
login entry 68
login entry 69
login entry 70
login entry 71
login entry 72
login entry 73
login entry 74
login entry 75
login entry 76
login entry 77
login entry 78
login entry 79
login entry 80
login entry 81
login entry 82
login entry 83
login entry 84
login entry 85
login entry 86
login entry 87
```

n. Display the contents of security.log starting from **line 101 until the end of the file**.

```
(prathiksha@kali) ~/practice33
$ tail -n +101 security.log
security event 101
security event 102
security event 103
security event 104
security event 105
security event 106
security event 107
security event 108
security event 109
security event 110
security event 111
security event 112
security event 113
security event 114
security event 115
security event 116
security event 117
security event 118
security event 119
security event 120
security event 121
security event 122
security event 123
security event 124
security event 125
security event 126
security event 127
security event 128
security event 129
security event 130
security event 131
security event 132
security event 133
security event 134
security event 135
security event 136
security event 137
security event 138
security event 139
security event 140
security event 141
```

- o. Find the total number of lines in login.log.

```
(prathiksha@kali)~/practice33  
$ wc -l login.log  
100 login.log
```

- p. Find the total number of lines in access.log.

```
(prathiksha@kali)~/practice33  
$ wc -l access.log  
100 access.log
```

- q. Find the total number of security events in security.log.

```
(prathiksha@kali)~/practice33  
$ wc -l security.log  
500 security.log
```

- r. Determine the total number of events recorded in incident.log.

```
(prathiksha@kali)~/practice33  
$ wc -l incident.log  
500 incident.log
```

- s. Count the number of words in login.log.

```
(prathiksha@kali)~/practice33  
$ wc -w login.log  
300 login.log
```

- t. Count the number of bytes in security.log.

```
(prathiksha@kali)~/practice33  
$ wc -c security.log  
9392 security.log
```

- u. Display the **line, word, and byte counts** of access.log.

```
(prathiksha@kali)~/practice33  
$ wc access.log  
100 300 1592 access.log
```

- v. Find the line counts of all four .log files using a single wc command.

```
(prathiksha@kali)~/practice33
$ wc -l login.log access.log security.log incident.log
100 login.log
100 access.log
500 security.log
500 incident.log
1200 total
```

- w. Display the first **20 entries** of login.log and, from those entries, display only the **last 5**.

```
(prathiksha@kali)~/practice33
$ head -20 login.log | tail -5
login entry 16
login entry 17
login entry 18
login entry 19
login entry 20
```

- x. Display the last **20 entries** of login.log and, from those entries, display only the **first 5**.

```
(prathiksha@kali)~/practice33
$ tail -20 login.log | head -5
login entry 81
login entry 82
login entry 83
login entry 84
login entry 85
```

- y. Display the first **30 entries** of access.log and then display only the **last 10 entries** from that output.

```
(prathiksha@kali)~/practice33
$ head -30 access.log | tail -10
access entry 21
access entry 22
access entry 23
access entry 24
access entry 25
access entry 26
access entry 27
access entry 28
access entry 29
access entry 30
```

- z. Display the last **30 entries** of access.log and then display only the **first 10 entries**.

```
(prathiksha@kali) ~/practice33  
$ tail -30 access.log | head -10  
access entry 71  
access entry 72  
access entry 73  
access entry 74  
access entry 75  
access entry 76  
access entry 77  
access entry 78  
access entry 79  
access entry 80
```

- aa. Display the first **50 events** from security.log and then display the **last 10 events** from those 50.

```
(prathiksha@kali) ~/practice33  
$ head -50 security.log | tail -10  
security event 41  
security event 42  
security event 43  
security event 44  
security event 45  
security event 46  
security event 47  
security event 48  
security event 49  
security event 50
```

- bb. Display the last **50 events** from security.log and then display the **first 10 events** from those 50.

```
(prathiksha@kali) ~/practice33  
$ tail -50 security.log | head -10  
security event 451  
security event 452  
security event 453  
security event 454  
security event 455  
security event 456  
security event 457  
security event 458  
security event 459  
security event 460
```

cc. Display entries from **line 11 to line 20** of login.log.

```
(prathiksha@kali)~/practice33
$ sed -n '11,20p' login.log
login entry 11
login entry 12
login entry 13
login entry 14
login entry 15
login entry 16
login entry 17
login entry 18
login entry 19
login entry 20
```

dd. Display entries from **line 21 to line 30** of login.log.

```
(prathiksha@kali)~/practice33
$ sed -n '21,30p' login.log
login entry 21
login entry 22
login entry 23
login entry 24
login entry 25
login entry 26
login entry 27
login entry 28
login entry 29
login entry 30
```

ee. Display entries from **line 11 to line 20** of access.log.

```
(prathiksha@kali)~/practice33
$ sed -n '11,20p' access.log
access entry 11
access entry 12
access entry 13
access entry 14
access entry 15
access entry 16
access entry 17
access entry 18
access entry 19
access entry 20
```

ff. Display entries from **line 31 to line 40** of access.log.

```
(prathiksha@kali)~/practice33  
$ sed -n '31,40p' access.log  
access entry 31  
access entry 32  
access entry 33  
access entry 34  
access entry 35  
access entry 36  
access entry 37  
access entry 38  
access entry 39  
access entry 40
```

gg. Display security events from **line 21 to line 30** of security.log.

```
(prathiksha@kali)~/practice33  
$ sed -n '21,30p' security.log  
security event 21  
security event 22  
security event 23  
security event 24  
security event 25  
security event 26  
security event 27  
security event 28  
security event 29  
security event 30
```

hh. Display security events from **line 51 to line 60** of security.log.

```
(prathiksha@kali)~/practice33  
$ sed -n '51,60p' security.log  
security event 51  
security event 52  
security event 53  
security event 54  
security event 55  
security event 56  
security event 57  
security event 58  
security event 59  
security event 60
```

ii. Display incident events from **line 101 to line 110** of incident.log.

```
(prathiksha@kali) [~/practice33]
$ sed -n '101,110p' incident.log
incident event 101
incident event 102
incident event 103
incident event 104
incident event 105
incident event 106
incident event 107
incident event 108
incident event 109
incident event 110
```

jj. Display incident events from **line 201 to line 220** of incident.log.

```
(prathiksha@kali) [~/practice33]
$ sed -n '201,220p' incident.log
incident event 201
incident event 202
incident event 203
incident event 204
incident event 205
incident event 206
incident event 207
incident event 208
incident event 209
incident event 210
incident event 211
incident event 212
incident event 213
incident event 214
incident event 215
incident event 216
incident event 217
incident event 218
incident event 219
incident event 220
```

kk. Display incident events from **line 491 to line 500**.

```
(prathiksha@kali) [~/practice33]
$ sed -n '491,500p' incident.log
incident event 491
incident event 492
incident event 493
incident event 494
incident event 495
incident event 496
incident event 497
incident event 498
incident event 499
incident event 500
```

ll. Take the first **20 entries** of login.log and count how many lines are displayed.

```
(prathiksha@kali)~/practice33  
$ head -20 login.log | wc -l  
20
```

mm. Take the last **15 entries** of access.log and count how many lines are displayed.

```
(prathiksha@kali)~/practice33  
$ tail -15 access.log | wc -l  
15
```

nn. Take the first **50 entries** of security.log and count the number of words in those entries.

```
(prathiksha@kali)~/practice33  
$ head -50 security.log | wc -w  
150
```

oo. Take the last **25 entries** of security.log and count the number of lines.

```
(prathiksha@kali)~/practice33  
$ tail -25 security.log | wc -l  
25
```

pp. Extract lines **21–30** from login.log and count the number of extracted lines.

```
(prathiksha@kali)~/practice33  
$ sed -e '21,30p' login.log | wc -l  
110
```

qq. Extract lines **51–60** from access.log and count the number of extracted lines.

```
(prathiksha@kali)~/practice33  
$ sed -n '51,60p' access.log | wc -l  
10
```

rr. Extract events **101–150** from incident.log and count how many events are displayed.

```
(prathiksha@kali)~/practice33  
$ sed -n '101,150p' incident.log | wc -l  
50
```

ss. Display the first **100 events** from incident.log, take the last **20 events** from that output, and count the number of lines.

```
(prathiksha@kali)-[~/practice33]  
$ head -100 incident.log | tail -20 | wc -l  
20
```